NanoObjectDetection Legacy Pipeline

Purpose and scope

The legacy implementation is the original Python 3.9-oriented package in `NanoObjectDetection/`. The complete example workflow is `/path/to/example/auswertung.py`; its settings are stored in `/path/to/example/parameter.json`.
The legacy code loads the complete image stack into memory and is therefore substantially more memory-intensive than the modern blockwise implementation.

The workflow estimates particle diffusion and hydrodynamic diameter from TIFF video data. It is especially intended for particles diffusing in a cylindrical fiber or channel. The supplied 100 fps example uses a nominal 20.6 micrometre channel, 0.3467 micrometres/pixel, water at 293.85 K, and 50 nm gold particles.

Running the legacy example

Edit `File.data_file_name`, `File.data_folder_name`, `File.json`, and the plot save paths in `/path/to/example/parameter.json` so they point to the local machine. `/path/to/example/auswertung.py` contains a original acquisition computer and must also be edited.

The script performs:

```
nd.CheckSystem.CheckAll(ParameterJsonFile)
settings = nd.handle_data.ReadJson(ParameterJsonFile)
rawframes_np = nd.handle_data.ReadData2Numpy(ParameterJsonFile)
rawframes_super = nd.handle_data.RoiAndSuperSampling(...)
rawframes_pre, rawframes_int, static_background = nd.PreProcessing.Main(...)
obj_all = nd.get_trajectorie.FindSpots(...)
traj_fixed = nd.get_trajectorie.Link(..., SearchFixedParticles=True)
obj_moving = nd.get_trajectorie.RemoveSpotsInNoGoAreas(...)
traj_moving = nd.get_trajectorie.Link(..., SearchFixedParticles=False)
traj_no_drift = nd.Drift.Main(...)
```

`CalcDiameter.Main2` is then called separately for `x`, `y`, and `2d`, and `nd.visualize.PlotDiameters` writes the corresponding diameter plots and CSVs.
An optional animation can be produced at the end of the example script.

Parameter file

The JSON is grouped into `Exp`, `Fiber`, `File`, `Logger`, `ROI`, `PreProcessing`, `Find`, `Link`, `StationaryObjects`, `Split`, `Drift`, `MSD`, `Time`, `PostProcessing`, `Plot`, and `Animation`.

Experiment and physical calibration

```
Key  |  Meaning
---  |  ---
`Exp.NA`  |  Numerical aperture used by localization-precision theory.
```

Exp.lambda	Illumination wavelength in nm.
Exp.fps	Acquisition rate in frames/s.
Exp.ExposureTime	Exposure duration in seconds; relevant to motion blur and precision models.
Exp.Microns_per_pixel	Image-plane calibration.
Exp.Temperature	Absolute temperature in kelvin.
Exp.Viscosity	Dynamic viscosity in the legacy internal unit convention, approximately 9.8466e-16 N*s/um^2 for the exa
Exp.Viscosity_auto	Obtain viscosity from the solvent model when enabled.
Exp.solvent	Solvent name used by automatic viscosity calculation.

The supplied viscosity corresponds to approximately 0.0009847 Pa s. Unit consistency is essential; an error here scales every diameter.

Fiber and geometry

Key	Meaning
Fiber.Mode	Optical/channel mode such as `Gaussian` or `Speckle`.
Fiber.Waist	Gaussian waist parameter where applicable.
Fiber.TubeDiameter_nm	Nominal channel diameter used in hindrance correction.
Fiber.Center_y_px	Optional calibrated channel centre for position-aware correction.

The default legacy example uses `TubeDiameter_nm = 20600`. The global legacy correction uses this value for every trajectory. Position-aware correction is optional and depends on the relevant settings and calibration.

Input, ROI, and preprocessing

Group/key	Meaning
File.data_type	Usually `tif_stack` for the example.
File.use_num_frame	`"all"` or a restricted frame count.
ROI.Apply	Enable ROI processing.
ROI.x_min/x_max, ROI.y_min/y_max	Spatial bounds.
ROI.frame_min/frame_max	Frame range limits.
PreProcessing.Remove_CameraOffset	Subtract a global camera offset estimate.
PreProcessing.Remove_Laserfluctuation	Correct illumination fluctuations; disabled in the reference example.
PreProcessing.Remove_StaticBackground	Estimate and subtract static background.
PreProcessing.AllowNonPosValues	Whether negative preprocessed values are retained.
PreProcessing.EnhanceSNR	Enable PSF/Gaussian-based preprocessing.
PreProcessing.KernelSize	PSF kernel size or automatic selection.
PreProcessing.ZNCC_min	Correlation threshold for applicable localization paths.

The camera offset is estimated globally. Static background removal uses temporal intensity statistics designed to reduce the influence of moving bright particles. These assumptions fail with strong illumination gradients, changing background structure, saturation, or substantial bleaching.

Detection and linking

Group/key	Meaning
Find.tp_diameter	Trackpy feature diameter in pixels; reference value 7.
Find.tp_minmass	Trackpy integrated-mass threshold; reference value 358.
Find.tp_separation	Minimum feature separation; reference value 20 pixels.
Find.tp_percentile	Detection percentile; reference value 20.
Link.Dark time	Maximum gap allowed during linking; reference value 1 frame.
Link.Max displacement	Maximum linking displacement; reference value 10 pixels.
Link.Max displacement fix	Smaller search range for stationary objects; reference value 0.3 pixels.
Link.Min tracking frames before drift	Minimum moving-track length before drift; reference value 10.
Link.Min_tracking_frames	Final minimum track length; reference value 5.

The numeric meaning of `tp_minmass` is specific to the legacy preprocessing and intensity representation. It should not be copied directly to the modern pipeline, where the reference equivalent was approximately 1000 after global

normalization.

Stationary objects, splitting, and drift

Group/key	Meaning
`StationaryObjects.Analyze fixed spots`	Run the stationary-object pass.
`StationaryObjects.Min distance to stationary object`	Exclusion radius around fixed spots.
`Split.Max_traj_length`	Split tracks longer than this length.
`Split.Max_traj_length_keep_tail`	Keep the final short fragment.
`Split.IntensityJump`	Split at intensity discontinuities.
`Split.Max rel median intensity step`	Relative intensity-jump threshold.
`Drift.Drift rolling window size`	Smoothing window for drift estimation.
`Drift.Min particle per block`	Minimum particles required for a drift block.
`Drift.Apply`	Enable drift correction.

The reference workflow first identifies fixed tracks, removes nearby moving detections, links moving particles, and then subtracts global drift. Fixed objects can also be contaminants; no-go filtering can remove valid particles near them.

MSD, diffusion, and diameter physics

For a trajectory coordinate x(t), the legacy code calculates:

```
MSD(tau) = average( (x(t+tau) - x(t))^2 )
```

For one coordinate:

```
MSD = 2*D*tau + b
```

and for two dimensions:

```
MSD = 4*D*tau + b
```

The slope is fitted over the configured lag range. The reference settings use `MSD.lagtimes_min = 1`, `MSD.lagtimes_max = 3`, while automatic lag selection can be enabled through `Amount lagtimes auto`. `Amount summands` is a minimum statistical-data requirement at each lag. `Estimate_X` selects the localization precision model used when optimizing lag selection.

The unconfined Stokes--Einstein inversion is:

```
d = kB*T/(3*pi*eta*D)
```

The legacy code can invert a hindrance factor using `MSD.EstimateHindranceFactor`. The hindrance polynomial is a Dechadilok--Deen-style approximation based on particle/channel diameter ratio. With position-aware correction enabled, `HindrancePositionAware` and the calibrated transverse centre are used to estimate a local available channel diameter.

Brownian-motion and post-processing checks

Key	Meaning
`MSD.CheckBrownianMotion`	Apply the Kolmogorov--Smirnov Brownian-motion screening.
`MSD.EvalOnlyLongestTraj`	Evaluate only the longest trajectory when enabled.
`PostProcessing.MinimalBrightness`	Exclude dim particles.

Key	Meaning
`PostProcessing.MaxDiameter`	Exclude diameters above a chosen limit; zero disables the limit.
`PostProcessing.ForceUltraUniform`	Apply an additional distribution-based uniformity filter.

The KS test is a statistical screen, not a physical proof of Brownian motion. The longest-track option changes the population being reported and should not be confused with length weighting.

Plot and export settings

Important `Plot` keys are:

Key	Meaning
`Plot.MSD_fit`	Display/save MSD fits.
`Plot.Histogramm`	Produce diameter histograms.
`Plot.Histogramm_abs_or_rel`	Unweighted (`abs`), trajectory-length-weighted (`rel`), or both.
`Plot.Histogramm_Fit_1_Particle`	Fit a distribution under the uniform-particle assumption.
`Plot.Histogramm_Bin_size_nm`	Histogram bin width.
`Plot.Histogramm_Bins`	Number of histogram bins where applicable.
`Plot.Histogramm_min/max`	Explicit diameter plot limits.
`Plot.DiamOverTraj`	Plot diameter against trajectory length.
`Plot.SaveFolder`	Output directory.
`Plot.save_data2csv`	Export numeric results as CSV.

The reference analysis writes files such as `sizes_df_lin_long.csv`, `Diameter_Histogramm-long-unweighted.csv`, `Diameter_Histogramm-long-weighted.csv`, and `DiameterOverTrajLength_long.csv`.

Important limitations

The legacy workflow loads the full TIFF stack and can exceed available RAM, especially during background calculation and multiprocessing. It also relies on machine-specific paths and older Trackpy/NumPy/Pandas behavior.

The MSD fit assumes approximately Brownian motion and a linear MSD regime. It does not fully model motion blur, localization uncertainty, anomalous diffusion, flow, wall chemistry, non-circular geometry, or polydispersity. Hindrance correction is an approximation and is only physically defensible when the channel geometry, particle size, viscosity, and position calibration are valid. Detection thresholds depend strongly on background subtraction and intensity scaling. Inspect trajectory and MSD diagnostics rather than relying on a single histogram median.

Reference 100 fps settings

The supplied `/path/to/example/parameter.json` uses `eval_dim = "x"` in the longitudinal reference result, 100 fps, 0.3467 micrometres/pixel, 293.85 K, a 20.6 micrometre channel, `tp_diameter = 7`, `tp_minmass = 358`, `tp_separation = 20`, maximum displacement 10 pixels, a 0.3-pixel stationary search range, 10 frames before drift, and 5 frames for final analysis.